Project Plan: Variational Inference for GARCH-Family Models

Summary in Plain English

Stock prices swing more on some days than others, and stormy periods tend to cluster, much like the weather. This "volatility" is what banks and investors need to forecast in order to manage risk. The standard forecasting model is called GARCH. Fitting it the Bayesian way gives not just a forecast but a measure of how sure we can be, yet the traditional Bayesian tool (MCMC) works by slow trial and error. This project tests whether a faster-in-principle machine-learning shortcut, Variational Inference (VI), gives the same answers. We re-run the Microsoft example from Magris and Iosifidis (2023), check our numbers against theirs, compare the forecasts on data the models have not seen, and repeat the test on the largest US companies. Technical terms are explained in the glossary (Section 6).

1. Introduction and Objective

The generalized autoregressive conditional heteroskedasticity (GARCH) framework is widely utilized in finance econometric analysis of volatility models as a method to model the time varying volatility of financial returns, with maximum likelihood estimation (MLE) being the most common method for estimating GARCH family models. However, Bayesian estimation of GARCH family models usually uses Markov Chain Monte Carlo (MCMC) algorithms that are computationally expensive and difficult to scale (Magris and Iosifidis, 2023).

The goal of this project is to implement and assess the Variational Inference (VI) procedure proposed by Magris and Iosifidis (2023) for estimating GARCH-family models. We test whether VI gives estimates and forecasts that are as accurate as MCMC and Quasi-Maximum Likelihood (QML), and we measure its computational cost against MCMC directly rather than assuming it.

2. Literature Overview

Bayesian machine learning has recently shown that Variational Inference (VI) is a scalable substitute to sampling techniques in high dimensional settings. VI has recast the problem of estimating posteriors into an optimization problem by approximating the true posterior with a family of tractable distributions (Blei et al., 2017).

Although VI has been fairly successful in the area of machine learning, there are few realised examples of its use within the field of financial econometrics, especially with reference to the GARCH family of models (Magris & Iosifidis, 2023). The paper undertakes to identify this gap and propose VI as a valid alternative for Bayesian estimation of volatility models. More specifically, the authors have demonstrated that Gaussian VI can be applied to GARCH-type models by simple transformations of parameters, thus allowing for the ability to optimize without constraints (Magris & Iosifidis, 2023).

Through a large-scale experiment on 488 S&P 500 constituents, the authors compare four VI optimisers (BBVI, QBVI, MGVB, EMGVB) with MCMC and QML across ARCH, GARCH, GJR-GARCH, EGARCH, GJR-EGARCH and FIGARCH models. They find that VI performs within about 1% of QML on in-sample and out-of-sample metrics (NLL,

RMSE, MAD, QLIK), and that the full-covariance optimisers (MGVB, EMGVB) closely match the MCMC posterior, while the diagonal ones (BBVI, QBVI) reach a slightly lower bound. The paper motivates VI by its scalability, but it does not report a run-time comparison with MCMC; we therefore measure this ourselves.

3. Methodology

In this project, a zero-mean GARCH(1,1) model (Bollerslev, 1986) is estimated under a Bayesian approach, with Variational Inference as the estimation method. The conditional variance is:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

The posterior is approximated by a Gaussian variational distribution, found by maximising the Evidence Lower Bound (ELBO), which is equivalent to minimising the Kullback–Leibler (KL) divergence from the variational distribution to the true posterior (Blei et al., 2017). Following Magris and Iosifidis (2023, Proposition 2), the constrained parameters are mapped to an unconstrained vector θ with a standard normal prior: $\omega = \exp(\theta\omega)$, $\alpha = f(\theta\alpha)$, $\beta = f(\theta\beta)(1 - \alpha)$, where f is the logistic function. This guarantees $\omega > 0$ and $\alpha + \beta < 1$ for every θ. Two VI algorithms are implemented: (i) black-box VI (BBVI, Ranganath et al., 2014), the paper's Algorithm 1, which estimates gradients from likelihood values only via the log-derivative trick, with a diagonal covariance; and (ii) a full-covariance Gaussian VI using the reparameterisation trick and automatic differentiation. Benchmarks are QML and a Random-Walk Metropolis MCMC sampler targeting the same posterior.

For the advanced component, the following extension is implemented, with a second one as a possible further step:

• Student-t innovations to capture the heavy tails of daily returns (Bollerslev, 1987), with $\nu = 2 + \exp(\theta\nu)$;

• A FIGARCH model to account for long-memory effects (Baillie, Bollerslev & Mikkelsen, 1996), as covered in the paper.

The code is written in Python (NumPy/Numba for the likelihood, QML and MCMC; PyTorch for the full-covariance VI) and executed in a Jupyter Notebook.

4. Data

The empirical analysis uses daily close-to-close log returns from Yahoo Finance. To allow a direct comparison with the paper's case study (Table II), the main dataset is Microsoft (MSFT) from 1 January 2018 to 30 June 2023 (1,383 trading days), the same window as the paper. As a smaller version of the paper's cross-sectional study, the analysis is repeated on the Dow Jones 30 constituents of the S&P 500. In line with the paper, the data are split chronologically into 75% for training and 25% for testing, and models are compared on NLL, RMSE, MAD and QLIK.

5. Expected Outcomes

- The project is expected to:
- Successfully implement a VI-based GARCH model
- Demonstrate whether VI provides comparable performance to MCMC and QML (Magris & Iosifidis, 2023)

- Illustrate volatility dynamics through estimated conditional variance, including posterior credible bands
- Compare the computational cost and practical trade-offs of VI and MCMC

6. Glossary of Key Terms

Volatility: how large the daily price swings are.

GARCH(1,1): a model that forecasts today's volatility from yesterday's price shock (α) and yesterday's volatility (β), plus a baseline (ω).

Stationarity: the condition $\alpha + \beta < 1$, which means shocks eventually fade and volatility cannot explode.

QML (Quasi-Maximum Likelihood): the classic method that finds the single best-fitting set of parameters.

Prior / posterior: what we believe about the parameters before and after seeing the data.

MCMC (Markov Chain Monte Carlo): the traditional Bayesian method, which approximates the posterior with a long random walk.

Variational Inference (VI): a machine-learning method that approximates the posterior with the closest simple (Gaussian) distribution.

ELBO: the score that VI maximises; higher means a closer approximation.

Train / test split: the model learns from the first 75% of the data and is evaluated on the unseen last 25%.

NLL, RMSE, MAD, QLIK: forecast-error measures; lower is better.

Student-t innovations: an alternative to the normal distribution that allows extreme days to occur more often ("fat tails").

7. References

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